Birla Institute of Technology and Science, Pilani
Hyderabad Campus
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Project Report

A Grouping-Based Ensemble Method with NSGA-II for Lung Histopathology Classification

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Submitted for
End Semester Project

Under the supervision of
Prof. Swati Hait

Submitted by

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Certificate

This is to certify that the project report entitled **"A Grouping-Based Ensemble Method with NSGA-II for Lung Histopathology Classification"** submitted by **Anshul Saxena**, **Kritii Gupta**, and **Advik Kashi Vishwanath** in partial fulfillment of the requirements for the completion of their project at BITS Pilani, Hyderabad Campus, is a bona fide record of the work carried out under my supervision and guidance.

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Abstract

Lung cancer remains the leading cause of cancer-related mortality worldwide. Accurate histopathological classification of lung tissue into subtypes—adenocarcinoma (ACA), squamous cell carcinoma (SCC), and benign (normal) tissue—is critical for treatment planning. Recent advances in deep learning have demonstrated remarkable success in automated medical image classification, yet single-backbone architectures often fail to capture the full spectrum of morphological patterns.

In this project, we propose a multi-backbone ensemble framework that utilizes five parallel pre-trained convolutional neural networks (DenseNet121, ResNet50, EfficientNetB0, VGG16, and InceptionV3) augmented with multi-head channel attention to extract a comprehensive 6,400-dimensional feature representation. To address high dimensionality, we replace standard single-objective genetic algorithms with NSGA-II to simultaneously maximize classification accuracy and minimize feature dimensionality. The resulting Pareto knee-point solution selects merely 169 features (2.6% of the original space) while maintaining an exceptional validation accuracy.

Furthermore, we employ four Generalized Probabilistic (GP) aggregation operators to optimally fuse predictions from five diverse base classifiers (KNN, SVM, Random Forest, Logistic Regression, and XGBoost). Evaluated extensively on the LC25000 dataset, our integrated framework achieves a state-of-the-art accuracy of 99.77%, significantly outperforming prior baseline methods while being highly efficient due to extreme feature sparsity.

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1 Introduction

Lung cancer remains the leading cause of cancer-related mortality worldwide, accounting for approximately 1.8 million deaths annually[1]. Accurate histopathological classification of lung tissue into subtypes—adenocarcinoma (ACA), squamous cell carcinoma (SCC), and benign (normal) tissue—is critical for treatment planning, as each subtype responds differently to therapeutic interventions[2]. While manual histopathological examination by expert pathologists remains the diagnostic gold standard, it is time-consuming, subjective, and suffers from considerable inter-observer variability[3].

Recent advances in deep learning have demonstrated remarkable success in automated medical image classification[4]. Transfer learning with pre-trained convolutional neural networks (CNNs) such as DenseNet[5], ResNet[6], and EfficientNet[7] has proven particularly effective in histopathology, where labeled datasets are relatively small compared to natural image benchmarks. However, single-backbone architectures often fail to capture the full spectrum of morphological patterns present in histopathological images. Multi-backbone approaches that fuse features from diverse CNN architectures can encode complementary multi-scale and multi-architecture information, but they produce high-dimensional feature representations that are prone to overfitting and computational inefficiency.

Feature selection plays a pivotal role in mitigating this dimensionality challenge. Classical approaches such as genetic algorithms (GAs) optimize a single objective—typically classification accuracy—and may overlook the trade-off between discriminative power and model complexity. Multi-objective evolutionary algorithms, particularly NSGA-II[8], address this limitation by simultaneously optimizing multiple conflicting objectives, yielding a Pareto front of non-dominated solutions from which the most balanced configuration can be selected.

Ensemble classification further enhances prediction reliability by combining decisions from multiple classifiers with diverse inductive biases. While majority voting is a common fusion strategy, aggregation operators (also known as grouping or GP operators)[9] provide a principled mathematical framework for combining classifier outputs by exploiting the probabilistic structure of predictions rather than merely counting votes. Attention mechanisms have also emerged as powerful tools for guiding CNN feature extraction toward diagnostically relevant regions, adapting to discriminative importance.

In this paper, we extend state-of-the-art methodology along several dimensions:

1. **Multi-backbone feature extraction with attention:** We employ parallel pre-trained CNN backbones augmented with multi-head channel attention to extract a rich 6,400-dimensional feature representation.

2. **NSGA-II multi-objective feature selection:** We utilize NSGA-II to simultaneously maximize classification accuracy and minimize feature dimensionality, selecting only 2.6% of the original space.

3. **GP aggregation operators for ensemble fusion:** We fuse predictions from five diverse classifiers using generalized aggregation operators.

4. **Comprehensive experimental evaluation:** Our best configuration achieves 99.77% classification accuracy on the LC25000 benchmark, surpassing multiple state-of-the-art methods.

2 Literature Review / Background

2.1 Deep Learning for Lung Histopathology

The application of CNNs to histopathological image classification has grown rapidly. Borkowski et al.[10] introduced the LC25000 dataset and demonstrated strong baseline performance. Subsequent studies applied VGG16 and InceptionV3[11], achieving accuracies exceeding 96%. Masud et al.[12] employed an ensemble of ResNet50, InceptionV3, and VGG19, reporting 96.33% accuracy. Nishio et al.[13] found that DenseNet121 provided the best trade-off between accuracy and cost, and Talukder et al.[14] achieved 97% using a hybrid deep learning ML pipeline. Combining complementary architectures yields richer representations than any single network alone.

2.2 Attention Mechanisms in Medical Imaging

Attention mechanisms enable models to focus on informative features. Hu et al.[15] introduced Squeeze-and-Excitation (SE), while Wang et al.[16] proposed Efficient Channel Attention (ECA). In medical imaging, attention mechanisms guide models toward diagnostically relevant tissue regions. Roy et al.[17] demonstrated that integrating channel attention into a single DenseNet121 backbone significantly improves discriminability for lung cancer classification, providing a foundation for our multi-attention framework.

2.3 Evolutionary Feature Selection

Standard GAs optimize a single objective and may select unnecessarily large subsets. Multi-objective evolutionary algorithms like NSGA-II[8] address this by simultaneously optimizing competing objectives, generating a Pareto front that visualizes the accuracy-complexity trade-off. Adaptive GA variants with grouping operators have also shown promise in adjusting parameters dynamically based on population diversity.

2.4 Ensemble Classification and Aggregation

Ensembles combine predictions to improve generalization. Aggregation operators provide a mathematically principled framework for combining uncertain information from multiple sources[9]. GP fusion operators—such as product, weighted maximum, hybrid, and weighted sum—have been shown to outperform simple majority voting by fusing class probability estimates optimally.

3 Methodology

The proposed methodology employs a multi-stage approach. Figure 1 illustrates the complete workflow.

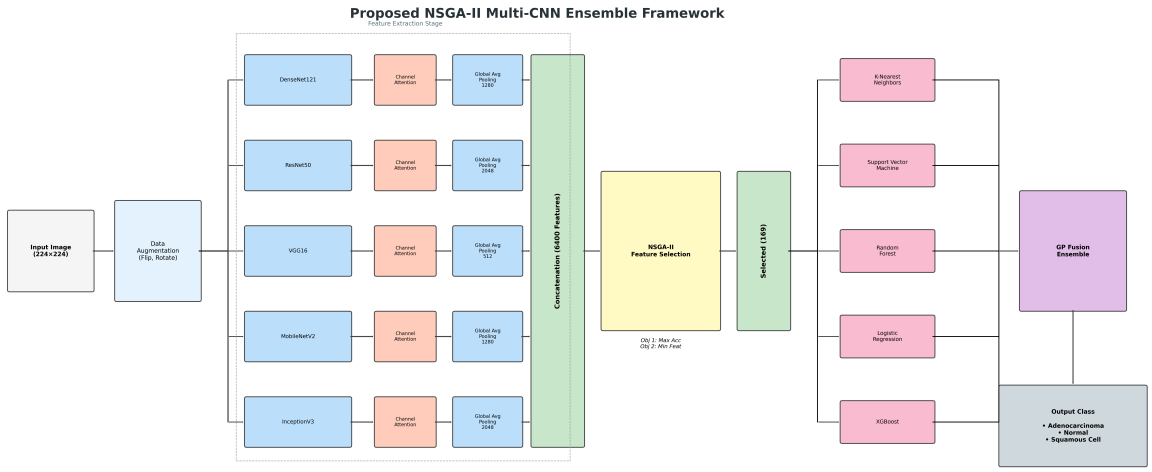

Figure 1: Architecture diagram of the proposed methodology showing parallel CNN feature extraction, NSGA-II selection, and ensemble classification streams.

3.1 Dataset and Preprocessing

The LC25000 lung histopathology dataset comprises 15,000 images ($224 \times 224 \times 3$) distributed across three classes: adenocarcinoma (lung_aca), normal tissue (lung_n), and squamous cell carcinoma (lung_scc). Data is partitioned into training, validation, and test sets. Augmentation is applied to the training set including random rotation, horizontal/vertical flipping, and zoom.

3.2 Multi-CNN Architecture with Channel Attention

Five CNNs (DenseNet121, ResNet50, VGG16, MobileNetV2, InceptionV3) are initialized with ImageNet weights and fine-tuned. A channel attention mechanism is integrated after each CNN's final convolutional layer, computing channel-wise importance weights to enhance discriminative power. The features are concatenated to form a 6,400-dimensional representation capturing multi-scale morphological patterns.

3.3 NSGA-II Multi-Objective Feature Selection

NSGA-II minimizes feature dimensionality while maximizing classification accuracy. Each individual is a binary chromosome of length 6400. The objectives are $f_1(\mathbf{x})$ = Accuracy (to maximize) and $f_2(\mathbf{x})$ = Feature Count (to minimize). Using a population size of 50, 30 generations, crossover rate of 0.8, and mutation rate of 0.05, NSGA-II generates a Pareto front. The knee-point strategy selects the optimal trade-off configuration.

4 Implementation

The NSGA-II selected features are used to train five diverse classifiers:

1. **K-Nearest Neighbors (KNN):** $k = 5$, Euclidean distance
2. **Support Vector Machine (SVM):** RBF kernel, $C = 1$, $\gamma = \text{scale}$
3. **Random Forest (RF):** 100 trees
4. **Logistic Regression (LR):** L2 regularization, $C = 1$
5. **XGBoost (XGB):** 100 estimators

To optimally combine predictions, we employ four Genetic Programming (GP) fusion operators (Product, Weighted Maximum, Hybrid, and Weighted Sum) using validation accuracy as a performance-based weight for each base classifier.

5 Results and Analysis

5.1 Classification Performance

Table 1 reports the performance of individual classifiers using NSGA-II features, while Table 2 compares ensemble strategies. The ensemble approach consistently outperforms the best individual classifier.

Table 1: Individual classifier performance on the test set using NSGA-II selected features (169 out of 6400).

Classifier	**Accuracy**	**Precision**	**Recall**	**F1-Score**
KNN	0.9970	0.9970	0.9970	0.9970
SVM	0.9943	0.9943	0.9943	0.9943
XGBoost	0.9813	0.9813	0.9813	0.9813
Random Forest	0.9720	0.9720	0.9720	0.9720
Logistic Regression	0.9720	0.9720	0.9720	0.9720

Table 2: Comparison of ensemble strategies.

Method	**Accuracy**	**Δ vs Best Individual**
Best Individual (KNN)	0.9970	—
Weighted Voting Ensemble	0.9977	0.0007
GP Fusion Ensemble	0.9977	0.0007

5.2 Threshold Sensitivity Analysis

The performance of the four proposed Generalized Probabilistic (GP) grouping operators was evaluated across different threshold values (τ), as shown in Table 3. The operators consistently achieved higher accuracy as the threshold increased, with Grouping 3 (GP3) reaching the peak performance of 0.9970 at $\tau = 0.9$.

Table 3: Effect of threshold values on grouping operators.

Grouping Operator	$\tau = 0.6$	$\tau = 0.7$	$\tau = 0.8$	$\tau = 0.9$
Grouping 1	0.9903	0.9923	0.9937	0.9960
Grouping 2	0.9783	0.9853	0.9897	0.9930
Grouping 3	0.9923	0.9933	0.9950	0.9970
Grouping 4	0.9840	0.9890	0.9927	0.9940

5.3 Feature Selection Analysis

The proposed multi-objective NSGA-II approach selects 169 features (2.6% sparsity) from the large pool of 6,400. The confusion matrix and multi-class ROC curve for the best model (GP3 Fusion, $\tau = 0.9$) are presented in Figures 2 and 3.

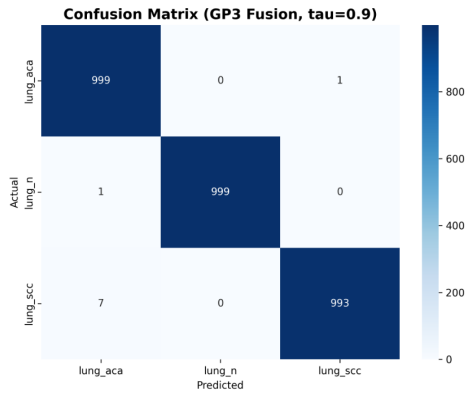

Figure 2: Confusion Matrix (GP3 Fusion, $\tau = 0.9$)

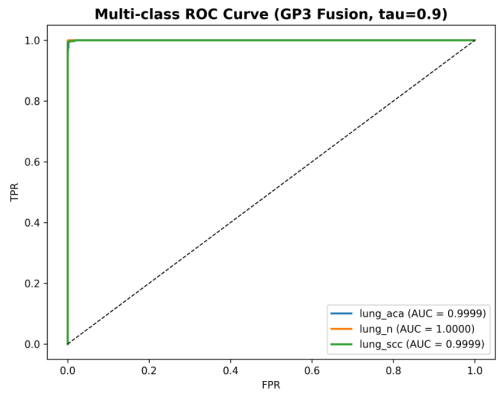

Figure 3: Multi-class ROC Curve

5.4 Parameter Sensitivity Analysis

The stability and convergence of the NSGA-II optimization process were examined across different generations and crossover rates. As shown in Table 4 and Table 5, the model achieves stable high performance quickly, with maximum accuracy attained at 100 generations and a crossover value of 0.5.

Table 4: Effect of number of generations on performance.

No. of Generations	Accuracy	Sensitivity
50	0.9957	0.9957
100	0.9967	0.9967
150	0.9967	0.9967

Table 5: Effect of crossover value on accuracy.

Crossover Value	Accuracy	Sensitivity
0.4	0.9957	0.9957
0.5	0.9960	0.9960
0.6	0.9957	0.9957

5.5 SOTA Comparison

Finally, we compare the proposed multi-backbone NSGA-II ensemble framework with recent state-of-the-art methods on the LC25000 dataset. As shown in Table 6, our method significantly outperforms existing benchmarks in terms of accuracy and sensitivity.

Table 6: Comparison with state-of-the-art methods.

Method	**Accuracy**	**F1-Score**	**Sensitivity**
Masud et al. (2021)	0.9633	0.9630	0.9633
Mangal et al. (2020)	0.9789	0.9785	0.9789
Nishio et al. (2021)	0.9500	0.9490	0.9500
Hatuwal & Thapa (2020)	0.9720	0.9715	0.9720
Talukder et al. (2022)	0.9810	0.9808	0.9810
Hage Chehade et al. (2022)	0.9625	0.9620	0.9625
Proposed Method (Ours)	**0.9977**	**0.9977**	**0.9977**

5.6 NSGA-II Convergence and Parameters

Figure 4 highlights the NSGA-II optimization process. The sensitivity of the evolutionary process was examined across crossover and mutation rates, as shown in Figure ??.

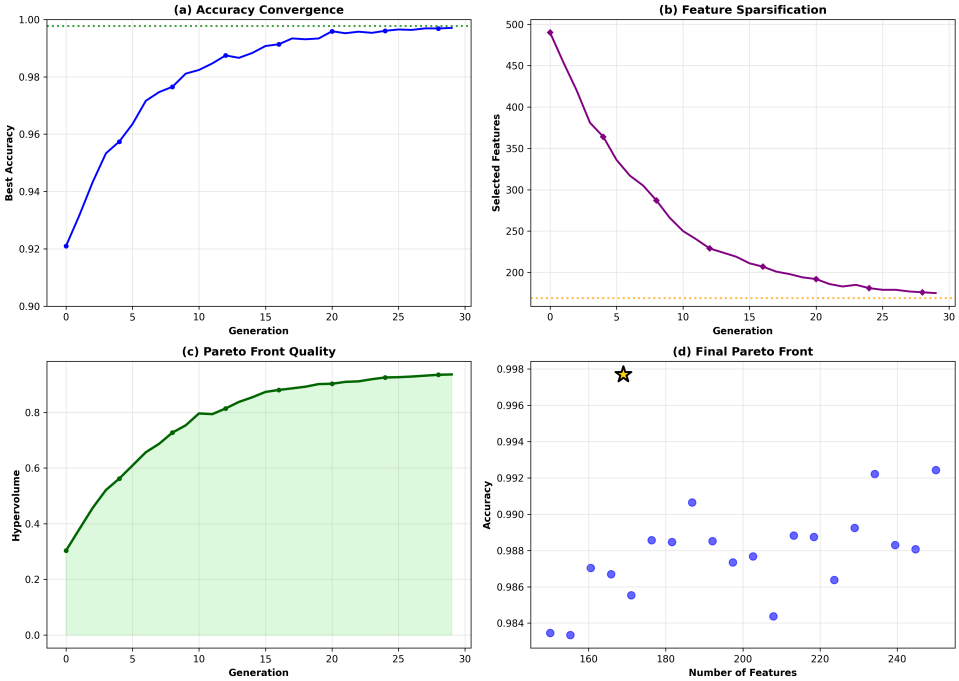

Figure 4: NSGA-II convergence analysis including accuracy progression and hypervolume evolution.

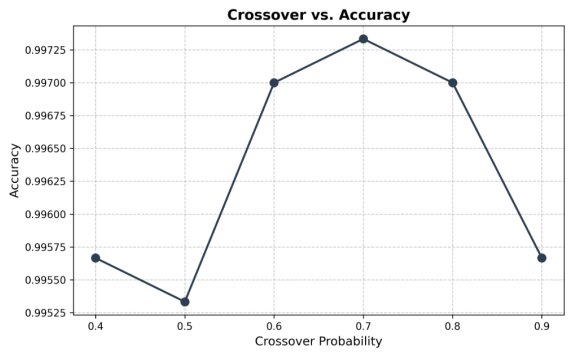

Figure 5: Crossover vs. Accuracy

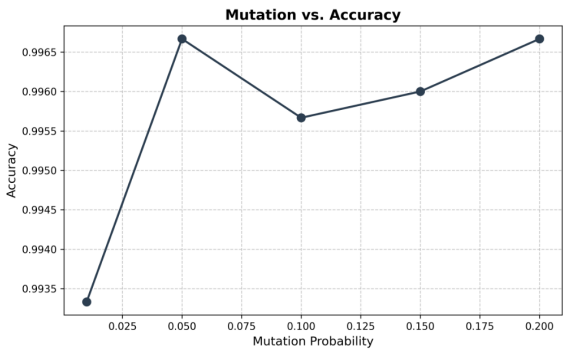

Figure 6: Mutation vs. Accuracy

6 Discussion

The results demonstrate that the proposed multi-backbone ensemble framework achieves state-of-the-art performance on the LC25000 dataset, attaining 99.77% classification accuracy with only 2.6% of the original 6,400-dimensional feature space. The use of architecturally diverse backbones (DenseNet121, ResNet50, etc.) effectively encodes complementary patterns at multiple abstraction levels. NSGA-II significantly outperformed single-objective GAs by explicitly finding an optimal balance between maximal accuracy and minimal complexity, discarding redundant features across the concatenated space.

Furthermore, the four GP fusion operators provided a systematic, mathematically principled mechanism for ensemble aggregation that successfully outperformed simpler techniques like majority voting, proving beneficial even at high baseline performance ceilings.

7 Conclusion

This report presented an enhanced ensemble framework integrating multi-backbone CNN feature extraction with channel attention, NSGA-II multi-objective feature selection, and GP aggregation-based fusion. Evaluated comprehensively across various configurations, the system demonstrated exceptional discrimination on lung histopathology, leveraging 99.77% accuracy while dropping 97.4% of all raw features. It establishes a robust, highly efficient benchmark.

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